



Extingo

An autonomous fire-detection and suppression system

Arduino Uno · multi-sensor detection · remote alerting · adaptive suppression

A smoke alarm can warn you. It cannot act for you.

Fires spread quickly. The highest-value response window is often the time before a person reaches the scene or emergency services arrive.

Fast

Fire growth can outpace human response

Passive

Most small-space systems only alert

Unattended

No one may be nearby to intervene

- + Traditional alarms detect smoke but do not identify direction.
- + They do not suppress the source or provide remote visibility.
- + Early heat rise can occur before a visible flame or heavy smoke.

Detect the threat, decide what it means, and respond immediately.

01

Autonomous detection

Continuously scans four directions using a two-servo mechanism and three independent sensing modes.

02

Autonomous response

Alerts the owner, locks aim, extracts smoke, sounds an alarm, and activates water suppression without manual intervention.

IR flame

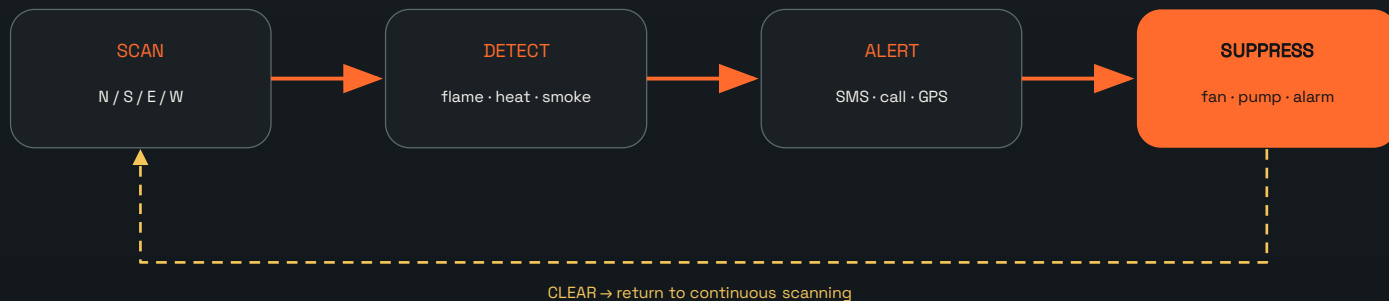
DHT22 heat

MQ2 smoke

SMS + call + GPS

PIR safety control

A closed-loop system from scan to clear.



Custom project flowchart can be inserted here

Each component has one job in the response chain.

MCU

Arduino Uno

Main controller and decision engine

IR

IR flame sensor

Primary directional detection

°C

DHT22

Heat level and rate-of-rise

SM

MQ2 sensor

Smoke and gas indication

↻

2× servos

Pan and tilt scanning

OLE
D**I2C OLED**

Live local status readout

GPS

GPS + GSM

SMS, call, and location

PIR

PIR sensor

Adaptive pump pressure

P

Pump + L298N

Water suppression drive

F

Fan + L298N

Smoke and heat extraction

S

ISD1820

Continuous audible alarm

Two perpendicular servos create four-direction coverage.

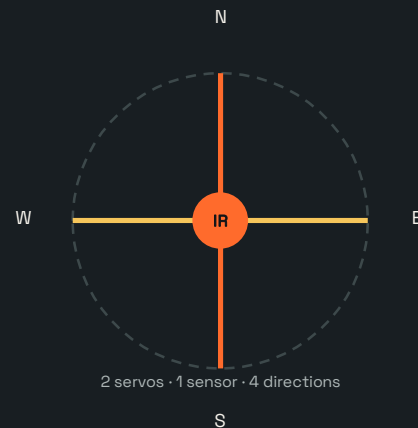
Lower servo — East / West

Sweeps the base horizontally across the room.

Upper servo — North / South

Mounted perpendicular on top, tilting the IR sensor vertically.

The IR sensor remains parallel to the upper servo, so the combined geometry maps cleanly to N / S / E / W.



Scanning never stops checking for fire while the servos move between positions.

Detection priority follows how a real fire develops.

01 IR flame sensor
Fastest and most direct evidence of flame

02 DHT22 heat sensor
Detects heat presence and sudden temperature rise

03 MQ2 smoke sensor
Useful confirmation, but smoke is not guaranteed early

“At the initial stage of a fire, flame and heat are almost always present — smoke is not guaranteed.”

Priority is deliberate, not arbitrary sensor triggering.

Once fire is detected, every second is assigned a task.

01**Lockaim**

Servo holds the detected direction.

**02****Alert owner**

SMS with direction + live location, then a call.

**03****Extract**

Exhaust fan turns on to move smoke and heat.

**04****Suppress**

Pump pressure is selected by PIR presence.

**05****Alarm + monitor**

ISD1820 loops; OLED refreshes every 3 seconds.

All actions stop automatically when the IR sensor reads clear — then Extingo returns to scanning.

The system can react to heat before a flame event.

 ΔT

Rate-of-rise trigger

DHT22 readings are compared over time. A sudden increase — for example, $\geq 2^{\circ}\text{C}/\text{sec}$ — triggers an early warning.

 $\geq 2^{\circ}\text{C}/\text{s}$

Example trigger threshold

SMS

Exact rate + temperature reading

A

Heat rises suddenly

DHT22 identifies abnormal change.

**B**

Fan starts automatically

Heat and smoke extraction begins.

**C**

Normal returns

Fan turns off and “All clear” SMS is sent.

Suppression pressure adapts to human presence.

PIR

Why this matters

Full-pressure water can be unsafe or wasteful if a person is standing near the fire — for example, someone already attempting a first response.

01

Person nearby

PIR detected → low pump pressure

02

Area clear

No PIR signal → high pump pressure

The robot is not only autonomous — it is designed to respond more safely

The owner stays informed even when they are not on site.

SMS**Direction-specific fire alert**

"Fire detected by Extingo — at [direction]" with live geo-location.

CALL**Immediate phone call**

Escalates the event beyond a notification banner.

 ΔT **Heat-rise warning**

Reports the exact temperature and rate-of-rise reading.

GPS**Live location**

Provides the robot's location to support a faster response.

OK**All-clear confirmation**

"All clear — Heat is now Low" after normal conditions return.

AUTO**No manual step**

Alerting begins automatically as part of the response sequence.

Extingo combines five response layers in one design.

01

Autonomous

Acts without waiting for human intervention.

02

Multi-sensor

Uses flame, heat, and smoke evidence.

03

Reasoned priority

Sensor order matches fire development.

04

Adaptive safety

PIR changes pump pressure near people.

05

Early warning

Rate-of-rise catches abnormal heat early.

Non-blocking firmware keeps detection, suppression, alerting, alarm, and OLED updates running concurrently.

Earlier detection. Faster response. Safer suppression.

Extingo turns fire safety from passive warning into active first response.

It senses the event, identifies the direction, alerts the owner, suppresses the fire, and returns to monitoring when conditions are clear.

Future scope

Mobility · larger-area coverage · app-based dashboard · stronger suppression hardware · advanced fire classification

Thank you